STUDY GUIDE

Revision Flashcards

Every revision card and quiz fact as a prompt and an answer

Cover the right-hand column, answer aloud, then check. Repeat the ones you miss after a day and again after a week.

Generated 10 October 2026 from the same lesson data that powers the website.
Every Java program in this guide was compiled with --release 17 and run against its expected output.

Revision flashcards

Core facts to remember

Question	Answer
Which operator gives the remainder in Java?	`%` — e.g. 17 % 5 = 2. The result takes the sign of the left operand: -7 % 3 = -1.
Why is `7 / 2` equal to 3 in Java?	Both operands are int, so Java performs integer division and drops the fraction. Use `7 / 2.0` for 3.5.
How do you compare two Strings by content?	`a.equals(b)` — `==` compares references, not characters.
What does `i++` vs `++i` do inside an expression?	`i++` yields the OLD value then increments; `++i` increments first and yields the new value.
int overflow: what is Integer.MAX_VALUE + 1?	Integer.MIN_VALUE (−2 147 483 648). Java integers wrap around silently.
What must every recursive method have?	A base case that stops the recursion, and a recursive step that moves toward it.
Pass-by-value for objects: what is copied?	The reference (the address). The called method can change the object, but cannot make the caller's variable point to a different object.
Difference between == and equals() for Integer objects?	`==` compares references (cached only for −128..127); `equals` compares values. Prefer equals.
What does a `break` do inside a switch?	Leaves the switch. Without it, execution falls through into the next case.
How do you get the last digit and drop the last digit of n?	Last digit: `n % 10`. Drop it: `n / 10`.
Stable vs unstable sorting?	A stable sort keeps equal elements in their original relative order (merge sort, insertion sort). Quick sort and selection sort are not stable.
What does "in-place" mean?	The algorithm uses O(1) extra memory beyond the input (it rearranges elements within the array).
Big-O vs Big-Omega vs Big-Theta?	O = upper bound (at most), Ω = lower bound (at least), Θ = tight bound (both).
Auxiliary space vs space complexity?	Auxiliary = extra temporary memory. Space complexity = auxiliary + the input itself.
Backtracking template?	choose → explore (recurse) → un-choose. Undo the choice so the next option starts from a clean state.
x & (x − 1) does what?	Clears the lowest set bit. x is a power of two exactly when x > 0 and x & (x−1) == 0.
Arithmetic vs logical right shift?	`>>` copies the sign bit (divides negatives correctly); `>>>` always shifts in 0.
Two's complement: how do you negate?	Flip every bit and add 1. −x = ~x + 1.

0.1 How Java Works: Source, Bytecode, JVM, JRE, JDK

Prompt	Answer
Recall this key fact from memory	.java → javac → .class (bytecode) → JVM → machine code.
Recall this key fact from memory	JDK ⊃ JRE ⊃ JVM.
Recall this key fact from memory	Write once, run anywhere: only the JVM differs per OS.
Recall this key fact from memory	JIT compiles hot bytecode to native code at run time.
What does the javac command produce?	bytecode (.class files) — javac compiles .java source into .class bytecode.
Which package must you install to COMPILE Java programs?	JDK — The JDK contains the compiler and tools.
Why can the same .class file run on Windows and macOS?	It is bytecode executed by a platform-specific JVM — The JVM hides the platform differences.
JIT compilation means...	compiling frequently used bytecode to machine code at run time — Hot code is compiled to native code just in time to speed it up.

0.2 The Java Landscape: History, Features, Editions, JVM Memory

Prompt	Answer
Recall this key fact from memory	Java SE = core; EE = enterprise servers; ME = embedded; JavaFX = GUIs.
Recall this key fact from memory	Stack: frames/locals. Heap: objects. GC frees unreachable objects.
Recall this key fact from memory	Features: OO, portable, robust, secure, multithreaded.
Where are Java objects stored?	heap — Objects live on the heap; references to them may live on the stack.
What does the garbage collector do?	frees objects that are no longer reachable — Automatic memory management.
Which edition is used for standard Java programs and algorithms?	Java SE — Java SE (Standard Edition).
A recursion that never stops usually ends with...	StackOverflowError — The call stack is exhausted.

0.3 Installing Java on macOS and Choosing an Editor

Prompt	Answer
Recall this key fact from memory	Install JDK 17+; verify java -version and javac -version.
Recall this key fact from memory	javac X.java → X.class; java X.
Recall this key fact from memory	File name = public class name.
Recall this key fact from memory	IDEs just call javac and java for you.
Which command compiles Hello.java?	javac Hello.java — javac is the compiler.
After compiling you run it with...	java Hello — Use the class name, without extension.
"javac: command not found" usually means...	no JDK (compiler) is installed or on the PATH — The compiler is missing from the PATH.
This course compiles programs for which Java language level?	Java 17 — Java 17 (LTS); newer JDKs compile with --release 17.

0.4 Your First Program: class, main, statements, comments

Prompt	Answer
Recall this key fact from memory	public class X {...} in X.java; main = public static void main(String[] args).
Recall this key fact from memory	Statements end with ; Blocks use { }.
Recall this key fact from memory	Comments: // line, /* block /, /* javadoc */.
Recall this key fact from memory	Case-sensitive; classes UpperCamelCase.
Where does a Java program start executing?	the main method — The JVM calls public static void main(String[] args).
What is printed? System.out.print("A"); System.out.print("B"); System.out.println("C"); System.out.println("D");	ABC D — print does not add a newline; println does.
The file for public class Counter must be named...	Counter.java — A public class must be in a file with exactly its name.

0.5 Compile-time, Runtime and Logic Errors

Prompt	Answer
Recall this key fact from memory	Compile-time: compiler refuses (syntax/type). Runtime: exception while running. Logic: wrong answer, no message.
Recall this key fact from memory	Read the first compiler error; use the line number.
Recall this key fact from memory	Stack trace: exception type, message, where.
Recall this key fact from memory	Check outputs against hand examples.
int x = "hello"; is...	a compile-time error — A type mismatch is found by the compiler.
The program prints 20 where 15 was expected, with no error message. This is a...	logic error — It runs fine but is wrong.
Which of these is a runtime error?	dividing an int by zero — ArithmeticException happens while running.
What does this print? int a = 7, b = 2; System.out.println(a / b);	3 — Integer division truncates the .5.
In "Hello.java:3: error: ';' expected" what does 3 mean?	line number — Line 3 of Hello.java.

0.6 What Is an Algorithm?

Prompt	Answer
Recall this key fact from memory	Algorithm = finite, unambiguous, effective steps from input to output.
Recall this key fact from memory	Write as English / pseudocode / code.
Recall this key fact from memory	Judge by correctness and efficiency.
Recall this key fact from memory	Families: search/sort, number theory, divide & conquer, greedy, backtracking, DP.
What is an algorithm?	A set of instructions to solve a problem — The slide's own MCQ: a set of well-defined instructions that solves a problem.
Which property is NOT required of an algorithm?	written in Java — It can be expressed in any notation.

Prompt	Answer
A correct algorithm that takes 10 years on real inputs is...	impractical: efficiency matters too — Correctness and efficiency are both needed.
To find the largest of n numbers you need at least...	n − 1 comparisons — Each of the other n − 1 numbers must be compared at least once.

1.1 Variables, Constants and Primitive Types

Prompt	Answer
Recall this key fact from memory	byte/short/int/long; float/double; char; boolean; String is a reference type.
Recall this key fact from memory	int default for whole numbers, double for decimals; L and f suffixes.
Recall this key fact from memory	final = constant; UPPER_SNAKE_CASE; variables lowerCamelCase.
Recall this key fact from memory	Initialise before use.
Which type should hold 5 000 000 000?	long — It exceeds the int limit of 2.1 billion.
Which is a valid variable name?	totalMarks — It cannot start with a digit, contain "-", or be a keyword.
What is printed? final int LIMIT = 10; int x = LIMIT * 2; System.out.println(x);	20 — LIMIT * 2 = 20.
Which of these is NOT a primitive type?	String — String is a class (reference type).

1.2 Type Conversion, Casting and Overflow

Prompt	Answer
Recall this key fact from memory	Widening automatic; narrowing needs a cast and may lose data.
Recall this key fact from memory	int/int is integer division; use 2.0 or a cast.
Recall this key fact from memory	Overflow wraps silently; widen BEFORE the operation; Math.xxxExact detects.
Recall this key fact from memory	char is a number: char + int → int.
What is printed? System.out.println(7 / 2 + 7 / 2.0);	6.5 — 3 + 3.5 = 6.5.
What is printed? int x = Integer.MAX_VALUE; x++; System.out.println(x < 0);	true — Overflow wraps to −2147483648.
Which statement computes 100000 × 100000 correctly?	long p = 100000L * 100000; — One operand is long, so the multiplication happens in long.
What is printed? char c = 'A'; c += 2; System.out.println(c);	C — Compound assignment includes an implicit cast.
(int) 9.99 equals...	9 — Casting truncates toward zero.

1.3 Arithmetic, Assignment and Increment Operators

Prompt	Answer
Recall this key fact from memory	/ on ints truncates; % is the remainder (sign of the left operand).
Recall this key fact from memory	x++ yields old value; ++x yields new.
Recall this key fact from memory	n % 10 = last digit; n / 10 = drop it.
Recall this key fact from memory	^ is XOR, not power.
What is printed? int x = 5; int y = x++ + ++x; System.out.println(y + " " + x);	12 7 — x++ yields 5 (x→6); ++x makes x 7 and yields 7. y = 12, x = 7.
What is -17 % 5? System.out.println(-17 % 5);	-2 — The remainder has the sign of the dividend.
Which expression gives the last digit of n?	n % 10 — n % 10.
What is printed? int n = 10; n += 5; n *= 2; n %= 7; System.out.println(n);	2 — 15 → 30 → 2.
What does 5 ^ 2 compute in Java?	7 — ^ is XOR: 101 ^ 010 = 111 = 7.

1.4 Relational, Logical and Ternary Operators; Precedence

Prompt	Answer
Recall this key fact from memory	< <= > >= == != give boolean; && || ! combine.
Recall this key fact from memory	&& and || short-circuit: right side may be skipped.
Recall this key fact from memory	cond ? a : b is an expression.
Recall this key fact from memory	Precedence: arithmetic > comparison > == > && > || > ?: > =; use parentheses.
What is printed? int a = 5, b = 0; System.out.println(b != 0 && a / b > 1);	false — b != 0 is false, so a / b is never evaluated (no exception).
What is printed? System.out.println(true || false && false);	true — && is evaluated before ||.
Which is the correct way to test 1 ≤ x ≤ 10?	1 <= x && x <= 10 — Two comparisons joined by &&.
What does 5 > 3 ? "yes" : "no" evaluate to?	"yes" — The condition is true, so the first value.

1.5 Bitwise Operators and Shifts

Prompt	Answer
Recall this key fact from memory	& AND, | OR, ^ XOR, ~ NOT, << >> >>>.
Recall this key fact from memory	mask = 1 << i; read (x>>i)&1; set |; clear & ~; toggle ^.
Recall this key fact from memory	x & (x−1) clears lowest set bit → power-of-two test, popcount.
Recall this key fact from memory	<< k = ×2^k; >> keeps sign; >>> fills zeros.
What is 12 & 10? System.out.println(12 & 10);	8 — 1100 & 1010 = 1000.
What is 5 ^ 3? System.out.println(5 ^ 3);	6 — 101 ^ 011 = 110.
Which expression sets bit 3 of x?	x | (1 << 3) — OR with the mask forces the bit to 1.
What is -16 >> 2? System.out.println(-16 >> 2);	-4 — Arithmetic shift divides by 4 keeping the sign.

Prompt	Answer
x & (x - 1) does what?	clears the lowest set bit — Subtracting 1 flips the lowest set bit and everything below it.

1.6 Output: print, println, printf and String Concatenation

Prompt	Answer
Recall this key fact from memory	print (same line), println (newline), printf (format, add %n).
Recall this key fact from memory	+ adds numbers, joins once a String is involved; left to right.
Recall this key fact from memory	%d %f %.2f %s %c %b %x %n; %5d width, %-5d left, %05d zeros, %,d groups.
Recall this key fact from memory	char prints as a char; char+int is int.
What is printed? System.out.println(1 + 2 + "3" + 4 + 5);	3345 — 1 + 2 = 3, then "3" joins → "33", then "4", "5" → 3345.
What is printed? char c = 'A'; System.out.println(c + 1);	66 — char + int is an int.
Which prints 3.14?	printf("%.2f", 3.14159) — %.2f formats a decimal with two digits.
Which method does NOT end the line?	print — print leaves the cursor on the same line.
System.err is used for...	error messages — The standard error stream.

1.7 Input: Scanner, BufferedReader and Command-Line Arguments

Prompt	Answer
Recall this key fact from memory	Scanner: nextInt/nextDouble/next/nextLine; hasNextXxx for EOF.
Recall this key fact from memory	After nextInt call nextLine once before reading a line.
Recall this key fact from memory	BufferedReader.readLine + parseInt for speed.
Recall this key fact from memory	args are Strings; check args.length.
After sc.nextInt() the next sc.nextLine() returns...	an empty string (the rest of the number's line) — The newline is still waiting.
java A 1 8 with System.out.println(args[0] + args[1]) prints... public static void main(String[] args) { System.out.println(args[0] + args[1]); }	18 — String concatenation.
Which class is best for reading 10^6 numbers fast?	BufferedReader — Buffered input is the fastest of the three.

1.8 Decision Making: if, else, switch

Prompt	Answer
Recall this key fact from memory	Condition must be boolean; always use braces.
Recall this key fact from memory	else-if ladder: first true wins — order matters.
Recall this key fact from memory	switch: break stops fall-through; default for the rest; types int/char/String/enum.
Recall this key fact from memory	Arrow switch (Java 14+) has no fall-through.

Prompt	Answer
What happens with int x = 10; if (x) { ... }?	compile-time error — incompatible types: int cannot be converted to boolean.
What is printed? int i = 9; switch (i) { case 0: System.out.println("zero"); break; case 1: System.out.println("one"); break; default: System.out.println("many"); }	many — No case matches 9, so default runs.
What is printed? int x = 2; switch (x) { case 1: System.out.print("a"); case 2: System.out.print("b"); case 3: System.out.print("c"); default: System.out.print("d"); }	bcd — Fall-through from case 2.
What does if (true) ; do?	nothing — ; is an empty statement — The empty statement is legal.

1.9 Loops: for, while, do-while, break and continue

Prompt	Answer
Recall this key fact from memory	for = known count; while = until condition; do-while = at least once.
Recall this key fact from memory	break exits, continue skips to next pass.
Recall this key fact from memory	Nested loops multiply: n × n.
Recall this key fact from memory	Check start, condition, update; trace with a table.
What is printed? int s = 0; for (int i = 1; i <= 4; i++) { if (i == 3) continue; s += i; } System.out.println(s);	7 — 1 + 2 + 4 = 7 (3 is skipped).
How many times does the body run? int count = 0; for (int i = 0; i < 3; i++) for (int j = 0; j < 4; j++) count++; System.out.println(count);	12 — 3 × 4.
A loop whose body must run at least once is best written as...	do-while — do-while tests after the body.
What is printed? int i = 1; while (i < 100) { i *= 3; } System.out.println(i);	243 — 1, 3, 9, 27, 81, 243 — stops when i ≥ 100.

1.10 Methods: Parameters, Return Values, Scope and Overloading

Prompt	Answer
Recall this key fact from memory	returnType name(params) { ... return value; }
Recall this key fact from memory	Arguments are copied (pass by value).
Recall this key fact from memory	Scope = enclosing braces.
Recall this key fact from memory	Overload = same name, different parameters.
What is printed? static void change(int n) { n = 99; } public static void main(String[] a) { int v = 5; change(v); System.out.println(v); }	5 — The method changed its own copy.
Which pair is a valid overload?	int f(int a) and int f(double a) — Different parameter types; return type alone does not distinguish.
A variable declared inside a for loop body is visible...	only inside that loop body — Scope is the enclosing block.

1.11 Tracing and Debugging Programs

Prompt	Answer
Recall this key fact from memory	Trace table: variables after every iteration.
Recall this key fact from memory	Debug loop: observe → hypothesise → inspect → fix → retest.
Recall this key fact from memory	Breakpoint, step over/into/out, variables pane.
Recall this key fact from memory	Test edges: empty, one, negatives, duplicates, max.
In a debugger, "step over" does what?	runs the current line without entering called methods — Step into enters methods; step over treats them as one step.
The best FIRST step when a program gives a wrong answer is...	trace a small failing input — Reproduce and observe.
What does FindTheBug.countEvensBuggy return for {2, 7, 4, 9, 6}? // loop starts at i = 1	2 — It skips a[0] = 2 and counts only 4 and 6.

2.1 One-Dimensional Arrays

Prompt	Answer
Recall this key fact from memory	Indices 0 ... length−1; a.length is a field.
Recall this key fact from memory	O(1) access; O(n) search/insert.
Recall this key fact from memory	= copies the reference; clone / Arrays.copyOf copies the data.
Recall this key fact from memory	Arrays.toString / sort / fill / equals.
What is printed? int[] a = {3, 1, 4, 1, 5}; System.out.println(a.length + " " + a[a.length - 2]);	5 1 — length is 5; index 3 holds 1.
What is printed? int[] a = {1, 2, 3}; int[] b = a; b[0] = 9; System.out.println(a[0]);	9 — b is an alias of a.
What does new int[3] contain?	three zeros — Numeric arrays are zero-initialised.
Which is the valid index range of an array of length n?	0 ... n−1 — Zero-based indexing.

2.2 Two-Dimensional Arrays (Matrices)

Prompt	Answer
Recall this key fact from memory	m.length = rows; m[i].length = columns.
Recall this key fact from memory	Nested loops: outer row, inner column.
Recall this key fact from memory	Diagonal i==j; anti-diagonal i+j==n−1; transpose t[j][i]=m[i][j].
Recall this key fact from memory	clone() is shallow for 2-D arrays.
What is printed? int[][] m = {{1, 2}, {3, 4}, {5, 6}}; System.out.println(m.length + " " + m[0].length + " " + m[2][1]);	3 2 6 — 3 rows, 2 columns, element in row 2 col 1.
The main diagonal of a square matrix is where...	i == j — Row index equals column index.
The time to read all cells of an R × C matrix is...	O(R·C) — Every cell once.
Which statement creates a 3-row jagged array?	new int[3][] — Only the first dimension is required.

2.3 Strings, StringBuilder and Characters

Prompt	Answer
Recall this key fact from memory	Immutable; methods return new Strings.
Recall this key fact from memory	Compare with equals, not ==. compareTo for order.
Recall this key fact from memory	substring(a, b) excludes b; indexOf → −1 if absent.
Recall this key fact from memory	StringBuilder for loops; c − 'a' maps letters to 0..25.
What is printed? String s = "hello"; System.out.println(s.substring(1, 3) + s.indexOf('l'));	el2 — substring(1,3) = "el"; first l is at index 2.
What is printed? String a = "hi"; String b = new String("hi"); System.out.println(a == b);	false — Different objects.
Which class should you use to build a long string in a loop?	StringBuilder — It avoids $O(n^2)$ copying.
What is printed? char c = 'd'; System.out.println(c - 'a');	3 — char arithmetic: 100 − 97.
Strings are...	immutable — Their contents never change after creation.

2.4 Number Systems, Binary and Two's Complement

Prompt	Answer
Recall this key fact from memory	Position weights: base^k. Binary 128 64 32 16 8 4 2 1.
Recall this key fact from memory	Decimal→base: repeated ÷, read remainders upwards. Base→decimal: Horner.
Recall this key fact from memory	1 hex digit = 4 bits.
Recall this key fact from memory	Two's complement: $-x = \sim x + 1$; n-bit range $-2^{n-1} \dots 2^{n-1}-1$.
What is 1011_2 in decimal? // 8 + 0 + 2 + 1	11 — 1·8 + 0·4 + 1·2 + 1·1.
What is the 8-bit two's complement pattern of −3? // 3 = 00000011	11111101 — Flip → 11111100, add 1 → 11111101.
One hexadecimal digit corresponds to how many bits?	4 — $16 = 2^4$.
The range of an 8-bit two's-complement integer is...	−128..127 — $-2^7 \dots 2^7 - 1$.
In repeated division by 2, the first remainder is...	the least significant bit — It is n mod 2.

2.5 Divisibility, GCD/LCM, Primes and Modular Arithmetic

Prompt	Answer
Recall this key fact from memory	Divisors pair (d, n/d): only look up to √n.
Recall this key fact from memory	gcd·lcm = a·b; coprime ⇔ gcd = 1.
Recall this key fact from memory	Modular + − × are safe to reduce at every step; use floorMod.
Recall this key fact from memory	powMod: square, multiply when exponent bit is 1.
gcd(18, 24) × lcm(18, 24) = ?	432 — It equals 18 × 24 = 432.

Prompt	Answer
What does Math.floorMod(-7, 3) return? System.out.println(Math.floorMod(-7, 3));	2 — −7 = −3·3 + 2.
To test whether n is prime by trial division you need to test divisors up to...	√n — A composite has a divisor ≤ √n.
What is (17 * 23) % 5? System.out.println((17 % 5) * (23 % 5) % 5);	1 — 2 · 3 = 6, 6 mod 5 = 1.

2.6 Recursion: Base Case, Recursive Case and the Call Stack

Prompt	Answer
Recall this key fact from memory	Base case + smaller recursive case.
Recall this key fact from memory	Stack frame per call; space = depth.
Recall this key fact from memory	Chain: n calls. Two branches: ~2^n. Halving: log n.
Recall this key fact from memory	Overlap → memoise.
What does f(4) return? static int f(int n) { if (n == 0) return 0; return n + f(n - 1); }	10 — 4 + 3 + 2 + 1 + 0.
What happens if a recursive method has no reachable base case?	StackOverflowError — Frames pile up until the stack is exhausted.
How many calls does naive fib(5) make (counting fib(5) itself)?	15 — calls(n) = calls(n−1) + calls(n−2) + 1: 1,1,3,5,9,15.
The SPACE used by add(n) = n + add(n−1) is...	O(n) — n stack frames at the deepest point.
Recursion tree with overlapping sub-problems suggests...	memoisation / dynamic programming — Cache each sub-answer once.

2.7 References, Pass-by-Value and Memory

Prompt	Answer
Recall this key fact from memory	Primitive variable = value; reference variable = address of a heap object.
Recall this key fact from memory	Pass-by-value: the reference is copied; the object can be modified, the caller's variable cannot be re-pointed.
Recall this key fact from memory	b = a aliases; clone/copyOf copies; deep copy for nested objects.
Recall this key fact from memory	null → NPE.
What is printed? static void f(int[] a) { a[0] = 7; a = new int[]{9}; } public static void main(String[] x) { int[] v = {1, 2}; f(v); System.out.println(v[0]); }	7 — The element change is seen; the reassignment is not.
Java passes arguments...	by value (references are copied) — Always by value; for objects the value is the reference.
Calling a method on a null reference throws...	NullPointerException — NullPointerException.
Where do objects live?	heap — The heap; references to them may be on the stack.

2.8 Boundary Conditions, Validation and Common Mistakes

Prompt	Answer
Recall this key fact from memory	Test: empty, one, max, negative, equal, sorted.
Recall this key fact from memory	Validate before use; try/catch for parsing.
Recall this key fact from memory	mid = lo + (hi − lo) / 2; use long for sums.
Recall this key fact from memory	Math.abs(MIN_VALUE) < 0.
What does this print? System.out.println(Math.abs(Integer.MIN_VALUE) < 0);	true — The absolute value overflows back to MIN_VALUE.
The safe way to compute the midpoint of lo and hi is...	lo + (hi - lo) / 2 — It cannot overflow when lo ≤ hi.
Which input is a boundary case for "find the maximum of an array"?	an empty array — There is no maximum of nothing.
Integer.parseInt("12.5") throws...	NumberFormatException — It is not a valid int.

3.1 Classes, Objects, Fields, Constructors and this

Prompt	Answer
Recall this key fact from memory	class = fields + constructors + methods; object = new Class(...).
Recall this key fact from memory	this = current object; this(...) chains constructors.
Recall this key fact from memory	Object variables are references; = aliases.
Recall this key fact from memory	Override toString for printing.
What is the difference between a class and an object?	a class is a blueprint; an object is an instance of it — new Student(...) builds an object from the Student class.
What is printed? class P { int v; P(int v) { this.v = v; } } P a = new P(1); P b = a; b.v = 5; System.out.println(a.v);	5 — a and b are the same object.
A constructor...	has the same name as the class and no return type — Written like a method without a return type.
What does this refer to inside a method?	the object the method was called on — The current object.

3.2 Encapsulation and Access Modifiers

Prompt	Answer
Recall this key fact from memory	private fields + public methods.
Recall this key fact from memory	private < package < protected < public.
Recall this key fact from memory	Validate in constructors/setters.
Recall this key fact from memory	Do not leak mutable internals.
Which modifier makes a member visible only inside its own class?	private — private.
Why make fields private?	so the class can enforce its rules and change internals later — Encapsulation protects invariants.

Prompt	Answer
A member with no modifier is visible...	in the same package — Package-private.

3.3 Inheritance, Overriding and Polymorphism

Prompt	Answer
Recall this key fact from memory	extends, super(...), @Override.
Recall this key fact from memory	Dynamic dispatch: actual type decides.
Recall this key fact from memory	instanceof + cast.
Recall this key fact from memory	is-a → inheritance; has-a → composition.
What is printed? class A { String f() { return "A"; } } class B extends A { String f() { return "B"; } } A x = new B(); System.out.println(x.f());	B — Dynamic dispatch uses the actual type.
Which keyword calls the parent constructor?	super — super(...).
Which relationship suggests composition rather than inheritance?	Car has an Engine — "has-a".
In the parent-child constructor chain, which runs first?	parent constructor — Parent first.

3.4 Abstract Classes and Interfaces

Prompt	Answer
Recall this key fact from memory	abstract class: state + partial code; cannot be instantiated.
Recall this key fact from memory	interface: contract; many per class; default methods.
Recall this key fact from memory	Program to the interface.
Recall this key fact from memory	Comparable<T>.compareTo → natural ordering.
Which can a class do?	implement several interfaces — One superclass, many interfaces.
An abstract method...	has no body and must be implemented by a concrete subclass — It declares a requirement.
To make Collections.sort(list) work for your class, implement...	Comparable<T> — compareTo defines the order.
Unrelated classes that share a capability are best modelled with...	an interface — An interface.

3.5 static, final and Initialisation Order

Prompt	Answer
Recall this key fact from memory	static = per class; instance = per object.
Recall this key fact from memory	final = assign once / no override / no extend.
Recall this key fact from memory	Order: static blocks → instance blocks → constructor.
Recall this key fact from memory	static methods have no this.
Why is main declared static?	the JVM calls it before any object exists — No object is needed to start the program.

Prompt	Answer
How many copies of a static field exist?	one per class — A single shared copy.
What is printed? class C { static int n; C() { n++; } } new C(); new C(); new C(); System.out.println(C.n);	3 — The static counter is shared.
A final variable of type int[]...	cannot be re-assigned but elements can change — final protects the reference.

3.6 Packages and Imports

Prompt	Answer
Recall this key fact from memory	package a.b.c; = folder a/b/c.
Recall this key fact from memory	import X; import pkg.*; import static ...;
Recall this key fact from memory	java.lang is automatic.
Recall this key fact from memory	Clash → fully qualified name.
Which package is imported automatically?	java.lang — java.lang.
Where must the package statement appear?	first in the file — It is the first statement (comments aside).
Does import java.util.*; import java.util.function.Function?	no — sub-packages are separate — Wildcards cover one package only.
The folder for package com.shop.billing; is...	com/shop/billing/ — Dots become folder separators.

3.7 Exceptions: try, catch, finally, throw and throws

Prompt	Answer
Recall this key fact from memory	try / catch(specific first) / finally.
Recall this key fact from memory	Checked: handle or declare; unchecked: fix the bug.
Recall this key fact from memory	throw new X("msg"); throws X.
Recall this key fact from memory	try-with-resources closes automatically.
Which is a checked exception?	IOException — IOException must be handled or declared.
What is printed? try { System.out.print("A"); int x = 1 / 0; System.out.print("B"); } catch (ArithmeticException e) { System.out.print("C"); } finally { System.out.print("D"); }	ACD — B is skipped; the catch and finally run.
What does throws do in a method signature?	declares that the method may throw a checked exception — It warns callers.
try-with-resources requires the resource to implement...	AutoCloseable — close() is called automatically.

3.8 Generics and Lists: ArrayList and LinkedList

Prompt	Answer
Recall this key fact from memory	List<E>: ArrayList (array) vs LinkedList (chain).
Recall this key fact from memory	Wrapper types for primitives; remove(int index) vs remove(Object).
Recall this key fact from memory	Modify during for-each → use removeIf / Iterator.
Recall this key fact from memory	ArrayList add amortised O(1); contains O(n).
What is printed? List<Integer> a = new ArrayList<>(List.of(5, 6, 7)); a.remove(1); System.out.println(a);	[5, 7] — remove(int) removes index 1 (the 6).
Which list is best for random access get(i)?	ArrayList — ArrayList is O(1).
Why is List<int> illegal?	generics need reference types; use Integer — Type arguments must be objects.
What is printed? Integer a = 1000, b = 1000; System.out.println(a == b);	false — Outside the −128..127 cache, boxed values are different objects.

3.9 Sets and Maps: HashSet, TreeSet, HashMap, TreeMap

Prompt	Answer
Recall this key fact from memory	Set = unique; Map = key → value.
Recall this key fact from memory	Hash O(1) avg unordered; Tree O(log n) sorted; LinkedHash insertion.
Recall this key fact from memory	Count: merge(k, 1, Integer::sum) / getOrDefault.
Recall this key fact from memory	Keys need equals + hashCode and should not change.
What does Set.add return when the element is already present?	false — false — the set is unchanged.
Which keeps keys sorted?	TreeMap — TreeMap, in O(log n).
Average cost of HashMap.get?	O(1) — Hashing jumps to the bucket.
What is printed? Map<String,Integer> m = new HashMap<>(); m.merge("a", 1, Integer::sum); m.merge("a", 1, Integer::sum); System.out.println(m.get("a"));	2 — merge sums existing and new values.

3.10 Queues, Deques, Stacks and PriorityQueue

Prompt	Answer
Recall this key fact from memory	Queue FIFO offer/poll/peek; Stack LIFO push/pop/peek on ArrayDeque.
Recall this key fact from memory	PriorityQueue = min-heap, O(log n) add/poll; reverseOrder for max.
Recall this key fact from memory	Brackets → stack; BFS → queue; top-k → heap.
Recall this key fact from memory	Avoid java.util.Stack.
What is printed? Deque<Integer> s = new ArrayDeque<>(); s.push(1); s.push(2); s.push(3); System.out.println(s.pop() + " " + s.peek());	3 2 — LIFO: 3 comes out first; 2 is now on top.
Which data structure is best for BFS?	queue — FIFO explores level by level.

Prompt	Answer
PriorityQueue<Integer>.poll() returns...	the smallest — Natural ordering is a min-heap.
Insert into a PriorityQueue costs...	O(log n) — Sift up through the heap.

3.11 Sorting with Comparable and Comparator

Prompt	Answer
Recall this key fact from memory	compare → negative / 0 / positive.
Recall this key fact from memory	Comparable: natural; Comparator: any order; thenComparing, reversed.
Recall this key fact from memory	Use Integer.compare, never a − b.
Recall this key fact from memory	Object sorts are stable, O(n log n).
A comparator returning a negative number means...	a goes before b — Negative: first argument first.
Why is (a, b) → a - b dangerous?	integer overflow — The subtraction can overflow.
A stable sort...	keeps equal elements in their original order — List.sort and Collections.sort are stable.
Time complexity of Collections.sort on n items?	O(n log n) — Merge-sort-like TimSort.

3.12 equals(), hashCode() and Immutability

Prompt	Answer
Recall this key fact from memory	Override equals AND hashCode with the same fields.
Recall this key fact from memory	Hash → bucket; equals → match inside bucket.
Recall this key fact from memory	Immutable: final class, private final fields, no setters, defensive copies.
Recall this key fact from memory	Use Objects.hash / Objects.equals.
If a.equals(b) is true then...	a.hashCode() == b.hashCode() must hold — That is the contract.
Which makes a class immutable?	private final fields and no setters — State cannot change after construction.
What does a HashSet<Point> print for size if equals is overridden but hashCode is not? set.add(new Point(1, 2)); set.add(new Point(1, 2)); System.out.println(set.size());	2 — Almost certainly: different hash codes put them in different buckets, so equals is never consulted.
Which of these classes is immutable?	String — String never changes.

4.1 Data Structures Overview: Arrays, Linked Lists, Stacks, Queues, Hash Tables

Prompt	Answer
Recall this key fact from memory	Array: O(1) index, O(n) insert. Linked list: O(1) head insert, O(n) index.
Recall this key fact from memory	Stack LIFO, queue FIFO, hash table O(1) average lookup.
Recall this key fact from memory	Reverse: save next, flip link, advance.
Recall this key fact from memory	Ring buffer: (head + size) % capacity.

Prompt	Answer
Which structure matches "undo"?	stack — The most recent action is undone first: LIFO.
Insert at the head of a singly linked list costs...	O(1) — Re-wire two references.
Which is best for BFS?	queue — FIFO explores by levels.

4.2 Trees, Graphs and Heaps

Prompt	Answer
Recall this key fact from memory	BST: left < node < right; inorder sorted; O(height).
Recall this key fact from memory	Graph: matrix $O(V^2)$ / list $O(V+E)$. BFS queue → shortest edges; DFS recursion → depth.
Recall this key fact from memory	Heap: parent $(i-1)/2$, children $2i+1$, $2i+2$; sift up/down O(log n).
Recall this key fact from memory	Mark visited when enqueuing.
Inorder traversal of a BST yields...	sorted keys — left, node, right.
Which data structure drives BFS?	queue — FIFO queue.
The children of heap index 3 (0-based) are at...	7 and 8 — $2\cdot3+1 = 7$ and $2\cdot3+2 = 8$.
Memory for an adjacency matrix of V vertices is...	$O(V^2)$ — One cell per vertex pair.

4.3 Searching: Linear and Binary Search

Prompt	Answer
Recall this key fact from memory	Linear O(n) any data; binary O(log n) sorted data.
Recall this key fact from memory	mid = lo + (hi − lo)/2; lo = mid+1 / hi = mid−1.
Recall this key fact from memory	lowerBound for duplicates and counting.
Recall this key fact from memory	Monotone test → binary search on the answer.
How many comparisons does binary search need (at most) for 1 000 000 sorted elements?	20 — $\lceil \log_2 10^6 \rceil = 20$.
Binary search requires the data to be...	sorted — Sorted order lets us discard a half.
The time complexity of binary search is...	O(log n) — The range halves each step.

4.4 Elementary Sorting: Bubble, Insertion, Selection; Stability

Prompt	Answer
Recall this key fact from memory	Bubble: adjacent swaps, early exit O(n) best. Insertion: shift into sorted prefix, adaptive. Selection: ≤ n−1 swaps, always n^2.
Recall this key fact from memory	All $O(n^2)$, O(1) space; bubble & insertion stable, selection not.
Recall this key fact from memory	Swaps = inversions.
Recall this key fact from memory	n small → insertion; else O(n log n).
Which of these sorts is NOT stable?	selection — Selection sort swaps across equal elements.
Insertion sort on an already sorted array takes...	O(n) — One comparison per element.

Prompt	Answer
How many swaps does bubble sort make on {3, 2, 1}? // adjacent swaps only	3 — 3 inversions → 3 swaps.
In-place means...	O(1) extra memory — Rearranges within the input.

4.5 Merge Sort and Divide and Conquer

Prompt	Answer
Recall this key fact from memory	Divide → recurse → merge; base case size ≤ 1.
Recall this key fact from memory	Merge two sorted ranges in O(n); take left on ties for stability.
Recall this key fact from memory	T(n) = 2T(n/2) + n → O(n log n); O(n) space.
Recall this key fact from memory	Inversions: add mid − i when taking from the right.
What is the time complexity of merge sort in the worst case?	O(n log n) — Always n log n.
Merge sort's extra space is...	O(n) — The temporary array.
The three steps of divide and conquer are...	divide, conquer, combine — Divide, conquer (recurse), combine.
How many inversions does {3, 1, 2} have? // pairs (3,1), (3,2)	2 — 3 > 1 and 3 > 2; 1 < 2.

4.6 Iteration versus Recursion

Prompt	Answer
Recall this key fact from memory	Loop O(1) space; recursion O(depth) stack.
Recall this key fact from memory	Branching recursion → memoise / bottom-up.
Recall this key fact from memory	Deep chains → loop or explicit stack.
Recall this key fact from memory	Trees, backtracking, D&C → recursion reads best.
Which uses O(1) extra space for factorial?	loop — Recursion needs a stack frame per level.
Memoisation turns naive Fibonacci from...	$O(2^n)$ to O(n) — Each value is computed once.
Which is best solved by recursion?	all subsets of a set — Branching choices form a tree.
What error may deep recursion cause?	StackOverflowError — The call stack is exhausted.

4.7 Time Complexity and Big-O Notation

Prompt	Answer
Recall this key fact from memory	Count basic operations vs n; drop constants & low-order terms.
Recall this key fact from memory	O upper, Ω lower, Θ tight.
Recall this key fact from memory	$1 < \log n < \sqrt{n} < n < n \log n < n^2 < n^3 < 2^n < n!$
Recall this key fact from memory	Sequential: add; nested: multiply.
$f(n) = 4n^2 + 100n + 7$ is...	$O(n^2)$ — The n^2 term dominates (n^3 is true but loose).
Which means "tight bound"?	Big-Θ — Θ is both an upper and a lower bound.

Prompt	Answer
Which is the fastest-growing?	2^n — Exponential beats every polynomial.
The time complexity of accessing an element of an array by index is...	O(1) — Constant: address arithmetic.

4.8 Algorithm Design Paradigms

Prompt	Answer
Recall this key fact from memory	Brute force (try all), D&C (split-solve-combine), greedy (local best), backtracking (choose-explore-undo), DP (store sub-answers).
Recall this key fact from memory	Greedy needs proof — coins {4,3,1} break it.
Recall this key fact from memory	DP: state + recurrence + base.
Recall this key fact from memory	Prune early in backtracking.
How many coins does the greedy method use for 30 with {25, 10, 5, 1}? // greedy takes 25, then 5	2 — 25 + 5.
A greedy algorithm is correct when...	the best local choice leads to a global optimum — Provability is the key.
Overlapping sub-problems suggest...	dynamic programming — Store and reuse answers.

4.9 Analysing Loops: Counting Iterations

Prompt	Answer
Recall this key fact from memory	i *= c or n /= c → log n. i·i ≤ n → √n.
Recall this key fact from memory	Nested multiply; dependent j ≤ i → $n^2/2$; j += i → n log n.
Recall this key fact from memory	break/early exit changes the count.
Recall this key fact from memory	Verify by measuring with a counter.
for (i = 1; i < n; i = i * 2) runs...	$\log_2$ n times — Geometric growth.
for (i=1;i<=n;i++) for (j=1;j<=i;j++) is...	$O(n^2)$ — n(n+1)/2.
for (i=1;i<n;i++) for (j=1;j<n;j=j+i) is...	O(n log n) — Harmonic series: n·H(n).
int fun(){ for(i=1;i<n;i++) for(j=1;j<sqrt(n);j++) ... } is...	O(n√n) — n outer × √n inner.
The loop while(n > 0) { ans += n % 10; n /= 10; } runs...	$\log_{10}$ n times (the number of digits) — One iteration per digit.

4.10 Analysing Recursion: Recurrence Relations

Prompt	Answer
Recall this key fact from memory	Write T(n); solve by unrolling or a recursion tree.
Recall this key fact from memory	T(n−1)+1 → n; T(n/2)+1 → log n; 2T(n/2)+n → n log n; $2T(n-1)+1 \to 2^n$.
Recall this key fact from memory	Master: compare n^d with n^log_b(a).
Recall this key fact from memory	Space = depth.
By the master theorem T(n) = 3T(n/2) + n is...	O(n^1.585) — Leaves dominate: $n^{\log_2 3}$.

Prompt	Answer
What is the recursion depth of merge sort?	log n — Halving.
Naive fib(n) makes about...	1.618^n calls — Exponential in the golden ratio.

4.11 Space Complexity and the Call Stack

Prompt	Answer
Recall this key fact from memory	Space = auxiliary + input; recursion = max depth × frame.
Recall this key fact from memory	Variables O(1); array O(n); matrix $O(n^2)$.
Recall this key fact from memory	fib naive: O(n) stack; halving recursion O(log n).
Recall this key fact from memory	Trade: memo O(n) memory for O(n) time.
Auxiliary space is...	the extra space used besides the input — Temporary memory.
A recursion that halves n each call has stack space...	O(log n) — Depth $\log_2 n$.
A 1000 × 1000 int matrix uses about...	4 MB — $10^6 \times 4$ bytes.

4.12 Best, Average and Worst Case

Prompt	Answer
Recall this key fact from memory	Case = which input; bound = O/Ω/Θ of its cost.
Recall this key fact from memory	Linear search 1 / n/2 / n; insertion sort n / n^2 / n^2; quick sort n log n / n log n / n^2.
Recall this key fact from memory	Worst case = guarantee.
Recall this key fact from memory	Randomise the pivot.
Worst-case input for insertion sort?	reverse sorted — Every element moves to the front.
Quick sort with the last element as pivot is $O(n^2)$ on...	already sorted data — Unbalanced partitions.
Which case is a guarantee for every input?	worst — The worst case bounds all inputs.
Merge sort's best, average and worst case are...	all O(n log n) — It always splits evenly.

5.1 Euclid's Algorithm (GCD)

Prompt	Answer
Recall this key fact from memory	gcd(a, b) = gcd(b, a mod b); gcd(a, 0) = a.
Recall this key fact from memory	Loop: r = a % b; a = b; b = r; stop when b == 0, answer is a.
Recall this key fact from memory	Time O(log min(a, b)); naive is O(min(a, b)).
Recall this key fact from memory	lcm(a, b) = a / gcd(a, b) * b (divide first to avoid overflow).
Recall this key fact from memory	Worst case: consecutive Fibonacci numbers.
What is gcd(48, 18)?	6 — 48 = 2·18 + 12, 18 = 1·12 + 6, 12 = 2·6 + 0 → gcd = 6.

Prompt	Answer
What does this program print? int a = 84, b = 36; while (b != 0) { int t = a % b; a = b; b = t; } System.out.println(a);	12 — 84 mod 36 = 12 → (36, 12); 36 mod 12 = 0 → (12, 0). a = 12.
What is the running time of Euclid's algorithm for gcd(a, b)?	O(log min(a, b)) — The remainder is more than halved every two steps, so only about $2 \cdot \log_2(\min)$ steps are needed.

5.2 Euler's Phi (Totient) Function

Prompt	Answer
Recall this key fact from memory	φ(n) = count of k in 1..n with gcd(k, n) = 1.
Recall this key fact from memory	$\varphi(n) = n \cdot \Pi(1 - 1/p)$ over DISTINCT primes p ∣ n.
Recall this key fact from memory	Code: for p ≤ √n strip p, result −= result/p; finally if n > 1: result −= result/n.
Recall this key fact from memory	O(√n) time, O(1) space. Prime: φ(p) = p − 1.
Recall this key fact from memory	φ(60) = 16 (not 32).
What is φ(17)?	16 — 17 is prime: every k from 1 to 16 is coprime to it. φ(p) = p − 1.
What is φ(60)?	16 — $60 = 2^2 \cdot 3 \cdot 5$ → 60 × 1/2 × 2/3 × 4/5 = 16. (The lecture slide says 32 — that is a mistake.)
What does phi(36) return for the optimized method? long result = 36, n = 36; for (long p = 2; p * p <= n; p++) { if (n % p == 0) { while (n % p == 0) n /= p; result -= result / p; } } if (n > 1) result -= result / n; System.out.println(result);	12 — $36 = 2^2 \cdot 3^2$ → 36 − 18 = 18 (p = 2); then 18 − 6 = 12 (p = 3).
Why is if (n > 1) needed after the loop?	A prime factor larger than √n can remain — The loop only searches p ≤ √n. A single bigger prime factor can survive as the remaining value of n.

5.3 Chinese Remainder Theorem

Prompt	Answer
Recall this key fact from memory	x ≡ r_i (mod m_i), m_i pairwise coprime → unique x modulo M = Πm_i.
Recall this key fact from memory	$x = \Sigma\, r_i \cdot M_i \cdot \text{inv}(M_i \bmod m_i)$, M_i = M / m_i, then x mod M.
Recall this key fact from memory	Slide example: x = 233 ≡ 23 (mod 105).
Recall this key fact from memory	Not coprime? Check consistency (gcd) first.
The smallest non-negative x with x ≡ 2 (mod 3), x ≡ 3 (mod 5), x ≡ 2 (mod 7) is...	23 — 233 mod 105 = 23 and 23 checks out for all three.
Which condition do the moduli need for the CRT formula?	Pairwise coprime — Pairwise coprime moduli make each M_i invertible modulo m_i.
What is printed? int x = 0; while (!(x % 4 == 1 && x % 5 == 2)) x++; System.out.println(x);	17 — Numbers ≡ 1 (mod 4): 1, 5, 9, 13, 17 → 17 % 5 = 2.
In the formula, what is M_i?	M / m_i — M_i = M / m_i is the product of all the other moduli.
The system x ≡ 1 (mod 6), x ≡ 2 (mod 4) has...	no solution — x ≡ 1 (mod 6) forces x odd; x ≡ 2 (mod 4) forces x even.

5.4 Combinations (nCr)

Prompt	Answer
Recall this key fact from memory	C(n,r) = n! / (r!(n−r)!)
Recall this key fact from memory	C(n,r) = C(n,n−r); C(n,0) = C(n,n) = 1
Recall this key fact from memory	Loop: result = result*(n−r+i)/i for i=1..r
Recall this key fact from memory	Pascal: C(n,r) = C(n−1,r−1) + C(n−1,r)
Recall this key fact from memory	Overflow: 13! > int, 21! > long
What is C(6, 2)?	15 — 6!/(2!·4!) = 720/(2·24) = 15.
How many ways to arrange (order matters) 2 of 6 items?	30 — 6 · 5 = 30. Dividing by 2! gives the 15 combinations.
What is printed? long result = 1; int n = 6, r = 2; for (int i = 1; i <= r; i++) result = result * (n - r + i) / i; System.out.println(result);	15 — i=1: 1·5/1 = 5; i=2: 5·6/2 = 15.
Why does fact(15) / (fact(2) * fact(13)) print a wrong answer in int arithmetic?	fact(15) overflows int — 15! ≈ 1.3×10^{12} > 2.1×10^{9}.
C(n, r) equals...	C(n, n−r) — Choosing r to include is the same as choosing n−r to leave out.

5.5 Karatsuba's Multiplication

Prompt	Answer
Recall this key fact from memory	x=a·10^m+b, y=c·10^m+d
Recall this key fact from memory	ac, bd, (a+b)(c+d); middle = third − ac − bd
Recall this key fact from memory	x·y = ac·10^(2m) + mid·10^m + bd
Recall this key fact from memory	T(n) = 3T(n/2) + O(n) → O(n^1.585)
Recall this key fact from memory	Base case: small factor → plain multiply
How many recursive multiplications does Karatsuba make per level?	3 — ac, bd and (a+b)(c+d).
Which expression equals ad + bc?	(a+b)(c+d) − ac − bd — Expanding (a+b)(c+d) = ac + ad + bc + bd.
The running time of Karatsuba is...	O(n^1.585) — T(n) = 3T(n/2) + O(n) gives n^($\log_2$ 3).
With a=12, b=34, c=56, d=78, what is (a+b)(c+d) − ac − bd? // ac = 672, bd = 2652 long abcd = 46 * 134; System.out.println(abcd - 672 - 2652);	2840 — 46 × 134 = 6164; 6164 − 672 − 2652 = 2840 = ad + bc.
Why do libraries use school-book multiplication for small numbers?	Karatsuba has overhead that dominates for small n — Recursion and extra additions cost more than 1-2 machine multiplications.

5.6 Booth's Algorithm (signed binary multiplication)

Prompt	Answer
Recall this key fact from memory	Pairs (Q0 Q-1): 10 → AC−M, 01 → AC+M, 00/11 → nothing; then arithmetic shift right.
Recall this key fact from memory	n cycles; product = AC:Q (2n bits).
Recall this key fact from memory	−M = ~M + 1.
Recall this key fact from memory	Run of 1s = 2^(k+1) − 2^j: one subtract, one add.
Recall this key fact from memory	Example: 7 × 3 = 0001 0101 = 21.
Booth's algorithm subtracts the multiplicand when (Q_0, Q_{-1}) is...	10 — "10" marks the start of a run of 1s.
What is the 4-bit two's complement of 0111 (7)? // flip: 1000, add 1	1001 — 1000 + 0001 = 1001 = −7.
An arithmetic right shift of 1100 (4-bit) gives...	1110 — The sign bit (1) is copied into the vacated left position.
What is 0111 × 0011 after 4 Booth cycles in AC:Q?	00010101 — 0001 0101 = 21 = 7 × 3.
Which multiplier makes Booth's algorithm most efficient?	11111111 — A single long run of 1s needs just one subtract (and the result is just a shift).

5.7 Simple Sieve of Eratosthenes

Prompt	Answer
Recall this key fact from memory	isPrime[0..n]=true (except 0,1)
Recall this key fact from memory	for p with p·p ≤ n and isPrime[p]: cross out p·p, p·p+p, ...
Recall this key fact from memory	Time O(n log log n), space O(n)
Recall this key fact from memory	Use for "all primes ≤ n" and many queries
In the sieve, why can the outer loop stop at p·p ≤ n?	Because every composite ≤ n has a prime factor ≤ √n — A composite m = a·b with a ≤ b has a ≤ √m ≤ √n.
Which numbers does p = 3 cross out for n = 30 (starting at p·p)? for (int m = 9; m <= 30; m += 3) System.out.print(m + " ");	9 12 15 18 21 24 27 30 — Multiples of 3 starting from 9.
Time complexity of the sieve of Eratosthenes?	O(n log log n) — Σ n/p over primes ≈ n log log n.
How many primes are there below 50?	15 — 2 3 5 7 11 13 17 19 23 29 31 37 41 43 47.

5.8 Segmented & Incremental Sieve

Prompt	Answer
Recall this key fact from memory	Base primes ≤ √R via simple sieve.
Recall this key fact from memory	window[i] ↔ L + i
Recall this key fact from memory	start = max(p·p, ⌈L/p⌉·p)
Recall this key fact from memory	skip numbers < 2
Recall this key fact from memory	Incremental: map composite → primes crossing it
To sieve [L, R] you need base primes up to...	√R — Every composite ≤ R has a prime factor ≤ √R.

Prompt	Answer
What is the first multiple of 7 that is ≥ 100 and ≥ 49? long low = 100, p = 7; System.out.println(Math.max(p * p, (low + p - 1) / p * p));	105 — (100 + 6) / 7 = 15; 15 · 7 = 105.
Why start marking at max(p·p, first multiple ≥ L)?	To avoid marking p itself and values below p·p — Smaller multiples were handled by smaller primes; p itself must stay unmarked.
Which statement describes an incremental sieve?	Produces primes one after another, remembering upcoming composites — It stores for each upcoming composite the primes that will cross it out.

5.9 Binary Palindrome

Prompt	Answer
Recall this key fact from memory	x & 1 = lowest bit; x >> 1 drops it; r << 1 | bit pushes it.
Recall this key fact from memory	Reverse the bits, compare with the original.
Recall this key fact from memory	O(log n) time, O(1) space.
Recall this key fact from memory	Even numbers (> 0) are never binary palindromes: they end in 0 but start with 1.
Which of these numbers is a binary palindrome?	9 (1001) — 1001 reads the same both ways.
What does this print? int x = 6, r = 0; while (x > 0) { r = (r << 1) | (x & 1); x >>= 1; } System.out.println(r);	3 — 6 = 110 → reversed 011 = 3.
What does r << 1 do to the number r?	multiplies by 2 — A left shift moves every bit one place to the left: multiplication by 2.
Time complexity of the bit-reversal palindrome test?	O(log n) — One iteration per bit; the number of bits is $\log_2 n$.

5.10 Swap Two Nibbles in a Byte

Prompt	Answer
Recall this key fact from memory	byte = 8 bits = 2 nibbles; one nibble = one hex digit.
Recall this key fact from memory	swap = ((b & 0xF0) >>> 4) | ((b & 0x0F) << 4)
Recall this key fact from memory	Mask with & 0xFF before shifting a Java byte.
Recall this key fact from memory	O(1).
How many bits are in a nibble?	4 — A nibble is half a byte: 4 bits.
What does this print (decimal)? int b = 0x3C; System.out.println(((b & 0xF0) >>> 4) | ((b & 0x0F) << 4));	195 — 0x3C → 0xC3 = 195.
Which mask keeps only the high nibble of a byte?	0xF0 — 0xF0 = 1111 0000.
Why write b & 0xFF for a Java byte b?	To remove sign extension and get 0–255 — Bytes are signed; & 0xFF keeps the low 8 bits as an unsigned value.

5.11 Longest Sequence of 1s (and after flipping a bit)

Prompt	Answer
Recall this key fact from memory	Scan bits: current++ on 1, reset on 0, best = max.
Recall this key fact from memory	Flip variant: candidate = previous + current + 1 at each 0; previous = current (unless next bit is 0 too).
Recall this key fact from memory	All ones (−1) → 32.
Recall this key fact from memory	O(32) time, O(1) space.
What does this print? int x = 0b0110_1110; int cur = 0, best = 0; for (int i = 0; i < 8; i++) { if ((x & 1) == 1) best = Math.max(best, ++cur); else cur = 0; x >>>= 1; } System.out.println(best);	3 — 01101110: runs of 1s are 3 (bits 1–3) and 2 (bits 5–6).
In the flip variant, which value is the candidate on a 0?	previous + current + 1 — Left run + flipped bit + right run.
The longest run of 1s after flipping one bit in -1 (all ones) is...	32 — All 32 bits are already 1; a 32-bit int cannot hold more.
What is the time complexity of the bit scan?	O(log n) (32 steps) — One step per bit; the number of bits is logarithmic in the value.

5.12 Josephus Problem

Prompt	Answer
Recall this key fact from memory	J(1) = 0; J(n) = (J(n−1) + k) mod n (0-based). Add 1 for 1-based.
Recall this key fact from memory	Simulation: index = (index + k − 1) % size; remove.
Recall this key fact from memory	k = 2: n = 2^m + r → 2r + 1.
Recall this key fact from memory	Iterative O(n) time, O(1) space; recursion depth n.
For N = 6 and k = 2, who survives?	5 — Order: 2, 4, 6, 3, 1 → survivor 5.
What does this print? int ans = 0; for (int people = 2; people <= 5; people++) ans = (ans + 2) % people; System.out.println(ans + 1);	3 — ans: 0 → (0+2)%2=0 → (0+2)%3=2 → (2+2)%4=0 → (0+2)%5=2; +1 = 3.
What is the time complexity of the iterative Josephus recurrence?	O(n) — One loop iteration per person.
For k = 2 and N = 20 (= 16 + 4) the survivor is...	9 — 2·4 + 1 = 9.

5.13 Block Swap Algorithm (Array Rotation)

Prompt	Answer
Recall this key fact from memory	Left rotate by d: result[i] = a[(i + d) % n]; use d mod n.
Recall this key fact from memory	AB → BA. Block swap shrinks like Euclid; three reversals is simplest.
Recall this key fact from memory	O(n) time, O(1) space.
Recall this key fact from memory	Right by d = left by n − d.
What is printed? int[] a = {1, 2, 3, 4, 5}; int d = 2, n = a.length; for (int i = 0; i < n; i++) System.out.print(a[(i + d) % n] + " ");	3 4 5 1 2 — Element i of the output is element (i + 2) mod 5 of the input.

Prompt	Answer
Which method rotates in O(n) time and O(1) extra space?	Three reversals — Reverse(A), reverse(B), reverse(all) uses only swaps.
Right rotation by 2 of a 7-element array equals left rotation by...	5 — n − d = 5.
The time complexity of the one-by-one rotation by d is...	O(n·d) — d passes of n element moves.
After reversal rotation of {1,2,3,4,5,6} left by 2, what is the 1st element? // reverse(0,1) → 2 1 3 4 5 6 // reverse(2,5) → 2 1 6 5 4 3 // reverse(0,5) → ?	3 — Final array: 3 4 5 6 1 2.

5.14 Leaders in an Array

Prompt	Answer
Recall this key fact from memory	Leader = strictly greater than all elements to its right; last element is always one.
Recall this key fact from memory	Scan from the right, keep max, a[i] > max → leader.
Recall this key fact from memory	O(n^2) brute force vs O(n) scan.
Recall this key fact from memory	Print in original order by reversing the stored list.
The leaders of {16, 17, 4, 3, 5, 2} are...	17 5 2 — 17 > all to its right; 5 > 2; 2 is last. 16 is smaller than 17.
The right-to-left scan runs in...	O(n) — One pass.
Is the last element of an array always a leader?	Yes — Nothing to its right can be larger.

5.15 Majority Element (Boyer–Moore Voting)

Prompt	Answer
Recall this key fact from memory	Majority: count > n/2.
Recall this key fact from memory	Boyer–Moore: candidate/count; count == 0 → adopt; same → ++; different → −−.
Recall this key fact from memory	ALWAYS verify the candidate with a second pass.
Recall this key fact from memory	O(n) time, O(1) space; > n/3 needs two candidates.
In an array of 9 elements, how many occurrences make a value the majority?	at least 5 — More than 9/2 = 4.5 → at least 5.
Why is the second pass of Boyer–Moore needed?	The candidate may not be a majority — If no majority exists, phase 1 still returns some value.
Which approach uses O(1) extra space and O(n) time?	Boyer–Moore — Voting needs only a candidate and a counter.
Sorting-based majority detection costs...	O(n log n) — The sort dominates.

5.16 Maximum Product Subarray

Prompt	Answer
Recall this key fact from memory	Keep max & min product ending here; swap if a[i] < 0.
Recall this key fact from memory	max = max(a[i], max·a[i]); min = min(a[i], min·a[i]); best = max(best, max).
Recall this key fact from memory	Zero resets; initialise from a[0], not 0.
Recall this key fact from memory	O(n) time, O(1) space; use long.
How many subarrays does an array of 5 elements have?	15 — n(n+1)/2 = 15.
What does the lesson's algorithm return for {−2, 3, −4}? // max/min product ending at each index	24 — (−2)·3·(−4) = 24.
Why do we keep the MINIMUM product ending at i?	A negative number can turn it into the maximum — Most negative × negative = largest positive.
Time complexity of the max/min algorithm?	O(n) — One pass with O(1) work per element.
Max product of {0, 2}? // a zero resets the running products	2 — The subarray {2} has product 2.

5.17 Maximum Equilibrium Sum

Prompt	Answer
Recall this key fact from memory	Equilibrium index: sum(left) == sum(right), a[i] excluded.
Recall this key fact from memory	right = total − left − a[i]; update left AFTER the test.
Recall this key fact from memory	Max equilibrium sum = max of left over equilibria (none → report none).
Recall this key fact from memory	O(n) time, O(1) space; use long.
For {1, 3, 5, 2, 2}, the equilibrium index (0-based) is...	2 — Left 1+3 = 4, right 2+2 = 4 at index 2.
What is right at index 1 of {4, 1, 2, 1}? long total = 8, left = 4; long right = total - left - 1; System.out.println(right);	3 — 8 − 4 − 1 = 3 (the elements 2 and 1).
How do you get the right sum in O(1)?	total − left − a[i] — Everything not on the left and not a[i] is on the right.
Time complexity of the single-pass solution?	O(n) — Two linear passes.

5.18 Maximum Sum of an Hourglass in a Matrix

Prompt	Answer
Recall this key fact from memory	Hourglass at (i, j): m[i][j..j+2], m[i+1][j+1], m[i+2][j..j+2].
Recall this key fact from memory	Positions: i in 0..R−3, j in 0..C−3 → (R−2)(C−2).
Recall this key fact from memory	Initial max = Integer.MIN_VALUE.
Recall this key fact from memory	O(R·C) time, O(1) space.
How many hourglasses fit in a 5×6 matrix?	12 — (5−2)(6−2) = 12.
Which cells belong to the hourglass anchored at (i, j)?	two full rows of 3 and the centre cell — Top row, centre of the middle row, bottom row.

Prompt	Answer
What is the hourglass sum of this 3×3? 1 2 3 / 4 5 6 / 7 8 9 // a b c / d / e f g	35 — 1+2+3+5+7+8+9 = 35 (the slide's own example).
Why initialise the maximum with Integer.MIN_VALUE?	All sums may be negative — Otherwise the result 0 would not correspond to any hourglass.

5.19 Toggle the Switch (100 Doors)

Prompt	Answer
Recall this key fact from memory	Monkey k toggles multiples of k; door n flips once per divisor.
Recall this key fact from memory	Odd number of divisors ⇔ perfect square.
Recall this key fact from memory	Open doors = $\lfloor \sqrt{n} \rfloor$.
Recall this key fact from memory	Simulation O(n log n); formula O(1).
After all 100 monkeys, how many doors are open?	10 — The perfect squares 1, 4, 9, ..., 100.
Door 36 is toggled how many times?	9 — Divisors of 36: 1, 2, 3, 4, 6, 9, 12, 18, 36 — nine (odd), so it is open.
Why is the number of divisors odd only for perfect squares?	Divisors pair up as (d, n/d) except when d = n/d — The unpaired divisor $\sqrt{n}$ makes the count odd.
The simulation's time complexity is...	O(n log n) — Σ n/k = n H(n).

5.20 Alice's Apple Tree (Minimum Garden Perimeter)

Prompt	Answer
Recall this key fact from memory	Ring r holds $12r^2$ apples; square of half-side r holds 2r(r+1)(2r+1).
Recall this key fact from memory	Perimeter = 8r.
Recall this key fact from memory	Smallest r with apples(r) ≥ needed → loop or binary search.
Recall this key fact from memory	Use long: 10^{15} needs r ≈ 503 968.
How many apples are in the ring at distance r = 2?	48 — $12 \cdot 2^2 = 48$ (and 12 + 48 = 60 inside the r = 2 square).
What does applesInside(2) return? static long applesInside(long r) { return 2 * r * (r + 1) * (2 * r + 1); }	60 — 2·2·3·5 = 60.
Why can binary search be used here?	apples(r) is monotonically increasing — More rings always mean more apples.
What is the perimeter of the square with half-side r?	8r — Side length 2r, four sides: 8r.

5.21 Weighted Substring (Smallest String of a Given Weight)

Prompt	Answer
Recall this key fact from memory	weight(n) = (n+1)·weight(n−1): 1, 3, 12, 60, 360 ...
Recall this key fact from memory	Greedy largest-first = fewest letters; sort for lexicographic order.
Recall this key fact from memory	Use long; stop generating weights at the target.
Recall this key fact from memory	Answer length is tiny (≤ ~160).
What is the weight of D?	60 — D = 4·C + C = 5·12 = 60.
Shortest string of weight 25?	ACC — 12 + 12 + 1 = 25 with just 3 letters.
What string does the lesson's method produce for weight 7? // weights A=1 B=3 C=12	ABB — Greedy takes B (3), B (3), then A (1): 3 + 3 + 1 = 7. Sorted: ABB.
Why does greedy work for these weights?	Each weight divides the next — Divisibility makes the coin system canonical.

5.22 Move Hyphen to the Beginning

Prompt	Answer
Recall this key fact from memory	indexOf, substring(a, b) = a..b−1, substring(a) = a..end.
Recall this key fact from memory	Strings are immutable; use StringBuilder for building.
Recall this key fact from memory	All hyphens: stable partition, right-to-left write pointer.
Recall this key fact from memory	O(n).
What is printed? String s = "ab-cd"; int i = s.indexOf('-'); System.out.println("-" + s.substring(0, i) + s.substring(i + 1));	-abcd — Before: "ab", after: "cd".
What does "face-prep".indexOf('x') return?	-1 — indexOf returns −1 when there is no match.
Why is building a long string with += in a loop slow?	Strings are immutable so every += copies the string — Each step creates a new String of increasing length.
Time complexity of the one-pass right scan?	O(n) — Every character is read once and written once.

5.23 Strobogrammatic Number

Prompt	Answer
Recall this key fact from memory	Rotations: 0↔0, 1↔1, 8↔8, 6↔9.
Recall this key fact from memory	Two pointers: rotate(s[l]) == s[r]; use l <= r.
Recall this key fact from memory	Middle digit ∈ {0, 1, 8}.
Recall this key fact from memory	Count for n digits: 4·5^(⌈n/2⌉−1)... (×3/÷5 for the odd middle).
Which number is strobogrammatic?	69 — 69 rotated by 180° is 69. 25 and 1962 contain 2 (no valid rotation); 962 contains 2 as well.
Which digits can be the MIDDLE digit of an odd-length strobogrammatic number?	0, 1, 8 — The middle digit must rotate into itself.
What does the check print for "88"? // pairs: (8,8) → rotate(8) = 8 ✓	true — 8 rotates to 8 and 88 reads the same.

Prompt	Answer
How many 2-digit strobogrammatic numbers (no leading zero) exist?	4 — 11, 69, 88, 96.

5.24 Lexicographically First Palindromic Rearrangement

Prompt	Answer
Recall this key fact from memory	Palindrome possible ⇔ ≤ 1 letter with odd count.
Recall this key fact from memory	Result = sorted half + odd letter + mirror.
Recall this key fact from memory	Normalise case; use TreeMap or int[26].
Recall this key fact from memory	O(n).
Which word can be rearranged into a palindrome?	level — level: l:2, e:2, v:1 — one odd count.
Smallest palindrome from "bbaac"? // counts a:2 b:2 c:1	abcba — front "ab", middle "c", back "ba".
Time complexity of the counting solution?	O(n) — One pass to count plus a pass over the alphabet.
In an even-length string, how many letters may have an odd count?	0 — The counts add up to an even number, so the number of odd counts is even; for a palindrome it must be ≤ 1, hence 0.

5.25 Manacher's Algorithm (Longest Palindromic Substring)

Prompt	Answer
Recall this key fact from memory	Pad with # → all palindromes odd. radius[i] = palindrome radius at i.
Recall this key fact from memory	Keep (center, right). If i < right: radius[i] = min(right − i, radius[2·center − i]).
Recall this key fact from memory	Extend while chars match; update center/right.
Recall this key fact from memory	O(n) time & space; original start = (c − r)/2, length r.
Why does Manacher's algorithm insert "#" between characters?	To make every palindrome odd-length — Even palindromes get a "#" centre.
The time complexity of Manacher's algorithm is...	O(n) — The right edge only moves forward.
The padded form of "abba" is... // insert # at the start, end, and between characters	#a#b#b#a# — n characters → 2n+1 characters.
The worst case input for centre expansion is...	aaaaaa — Every centre expands to the borders.

5.26 Natural Sort Order

Prompt	Answer
Recall this key fact from memory	Split into digit / non-digit chunks.
Recall this key fact from memory	Number chunks → numeric compare (BigInteger); text → compareToIgnoreCase.
Recall this key fact from memory	First difference decides; fewer chunks first.
Recall this key fact from memory	Add tie-breakers so the order is total.

Prompt	Answer
Plain String ordering puts which first?	file10 — Compare "1" with "2": 1 < 2, so file10 comes first.
How should two digit chunks "07" and "7" compare in natural order?	equal numerically (then tie-break) — Numerically they are both 7; add a tie-breaker for a deterministic order.
What does a Comparator return when a should come before b?	a negative number — Negative, zero, positive: less, equal, greater.
List.sort in Java is...	stable — It uses TimSort, a stable sort.

5.27 Maneuvering (Counting Grid Paths)

Prompt	Answer
Recall this key fact from memory	paths(m, n) = paths(m−1, n) + paths(m, n−1); first row/col = 1.
Recall this key fact from memory	Recursion is exponential; DP is O(m·n).
Recall this key fact from memory	Closed form: C(m+n−2, m−1).
Recall this key fact from memory	Obstacle ⇒ 0; use long / modulus.
Number of right/down paths in a 4 × 4 grid of cells?	20 — C(6, 3) = 20.
How many paths does this return? int[][] dp = new int[3][3]; for (int i = 0; i < 3; i++) for (int j = 0; j < 3; j++) dp[i][j] = (i == 0 || j == 0) ? 1 : dp[i-1][j] + dp[i][j-1]; System.out.println(dp[2][2]);	6 — Rows: 1 1 1 / 1 2 3 / 1 3 6.
Why is the recursive solution slow?	It recomputes the same sub-grids many times — Overlapping sub-problems → exponential calls.
Time complexity of the DP table solution?	O(m·n) — Each cell is filled once.

5.28 Sorted Distinct Permutations

Prompt	Answer
Recall this key fact from memory	Distinct permutations = n! / Π(c_i!).
Recall this key fact from memory	Next permutation: pivot, swap with next larger on the right, reverse suffix.
Recall this key fact from memory	Count-based backtracking: letters in alphabetical order, choose/explore/un-choose.
Recall this key fact from memory	Swap-based generation is NOT lexicographic.
How many distinct permutations does "AABB" have?	6 — 4!/(2!·2!) = 6.
What follows "ACB" in lexicographic order of "ABC"? // permutations of A, B, C	BAC — Pivot is A (A < C); smallest larger letter to its right is B; swap → BCA, reverse the tail "CA" → "AC" → BAC.
Which statement about the slide's swap-based generator is true?	It prints every distinct permutation once, but not in sorted order — The HashSet prevents duplicates; swapping disturbs the order.
Time to print all permutations of 8 distinct letters is...	O(8·8!) — 8! permutations × O(8) to build/print each.

5.29 N-Queens (Backtracking)

Prompt	Answer
Recall this key fact from memory	One queen per row; try columns left→right.
Recall this key fact from memory	choose → explore → un-choose.
Recall this key fact from memory	Conflict: same column, or row − col equal, or row + col equal.
Recall this key fact from memory	Counts: 1,0,0,2,10,4,40,92,352,724 for N = 1…10. O(N!) worst case.
How many solutions does the 4-queens problem have?	2 — Two mirror-image solutions.
Two queens at (1, 2) and (3, 4) attack each other because…	same diagonal — $\|\Delta row\| = \|\Delta col\| = 2$.
What does "backtracking" mean here?	Undoing the last choice and trying the next option — Return to the previous decision and change it.
Which strategy fits "find all arrangements that satisfy constraints"?	Backtracking — Systematic search with pruning.

5.30 Maze Solving (Rat in a Maze)

Prompt	Answer
Recall this key fact from memory	Maze = grid; neighbours via dr/dc arrays; bounds before wall test.
Recall this key fact from memory	DFS: visited, recurse, mark route when success returns true, backtrack on failure.
Recall this key fact from memory	BFS (queue) = shortest path.
Recall this key fact from memory	O(R·C).
Why does a 4-direction maze solver need a visited array?	To avoid walking in circles forever — Without it the search can bounce between two cells indefinitely.
Which algorithm guarantees the SHORTEST path in an unweighted maze?	BFS — BFS explores in rings of increasing distance.
What does sol[x][y] = 0; after both recursive calls do?	Un-mark a dead-end cell (backtrack) — The cell is not on any path to the exit.
DFS on an R × C maze with a visited array runs in…	O(R·C) — Each cell is expanded once.

5.31 Hamiltonian Cycle

Prompt	Answer
Recall this key fact from memory	Hamiltonian cycle: every vertex once, back to start.
Recall this key fact from memory	path[0] = 0; place vertices with edge from previous & unused; close with edge to path[0].
Recall this key fact from memory	choose / explore / reset to −1.
Recall this key fact from memory	O(V!) — NP-complete.
A Hamiltonian cycle visits…	every vertex exactly once and returns to the start — Edges once = Eulerian; vertices once = Hamiltonian.

Prompt	Answer
Why can the search always start at vertex 0?	A cycle can be rotated to start anywhere — Any cycle through all vertices contains vertex 0.
Time complexity of the backtracking search?	O(V!) — Up to (V−1)! orderings.
A vertex with only one neighbour makes a Hamiltonian cycle…	impossible — A cycle needs two edges at every vertex.

5.32 Knight's Tour & Warnsdorff's Rule

Prompt	Answer
Recall this key fact from memory	Knight: 8 moves (±1,±2),(±2,±1).
Recall this key fact from memory	Warnsdorff: go to the unvisited neighbour with the fewest unvisited neighbours.
Recall this key fact from memory	Heuristic — can fail; tie-break order matters.
Recall this key fact from memory	Backtracking exponential; heuristic $O(N^2)$.
How many moves can a knight in the middle of a large board make?	8 — Eight L-shaped jumps.
Warnsdorff's rule chooses the next square with…	the fewest onward moves — Visit the most constrained square first.
Is Warnsdorff's rule guaranteed to find a tour?	No, it is a heuristic — It can get stuck; check the result.
The heuristic's running time on an N × N board is…	$O(N^2)$ — N^2 moves with constant work each.

5.33 Selection Sort

Prompt	Answer
Recall this key fact from memory	Find minimum of the unsorted part, swap to the front, repeat.
Recall this key fact from memory	$O(n^2)$ always; ≤ n−1 swaps; in place; not stable.
Recall this key fact from memory	After pass i: first i+1 elements final.
What is the array after the 2nd pass? int[] a = {29, 10, 14, 37, 13}; // selection sort	10 13 14 37 29 — Pass 1: swap 10 with 29 → 10 29 14 37 13. Pass 2: min of rest is 13 → swap with 29 → 10 13 14 37 29.
How many comparisons does selection sort make on 6 elements, whatever their order?	15 — 5 + 4 + 3 + 2 + 1 = 15.
Which property is TRUE of selection sort?	At most n−1 swaps — One swap per pass at most.
Best-case time complexity of selection sort?	$O(n^2)$ — It always scans the entire unsorted part.

5.34 Quick Sort

Prompt	Answer
Recall this key fact from memory	Pivot → partition (≤ left, > right) → recurse both sides.
Recall this key fact from memory	Lomuto: i = low−1; scan j; if a[j] ≤ pivot {i++; swap}; swap pivot to i+1.
Recall this key fact from memory	Average O(n log n), worst O(n^2), space O(log n), in place, unstable.
Recall this key fact from memory	Quickselect: recurse on one side only.
After one partition, the pivot is...	in its final sorted position — Everything left is ≤ and everything right is >.
Worst-case time of quick sort?	O(n^2) — Highly unbalanced partitions.
What does partition return for this call? int[] a = {3, 8, 2, 5, 1, 4, 7, 6}; int p = partition(a, 0, 7); // pivot = a[7] = 6 System.out.println(p);	5 — Five elements (3, 2, 5, 1, 4) are ≤ 6 so the pivot goes to index 5.
Which strategy is used by quick sort?	Divide and conquer — Partition, then solve the two halves independently.
Is quick sort stable?	No — Partitioning swaps distant elements, so equal keys may be reordered.